

Coding Best Practices Activity

Fortune Teller (bad version vs good version)

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<https://github.com/dsamanta12344/coding-best-practices-activity>

What I made

The assignment said we could do a calculator or fortune teller or whatever as long as it had at least 3 functions, so I did a fortune teller. It asks for your name, birth month, and a lucky number and then prints a love fortune, a career one, and a luck one.

There's two python files. `fortune_teller_issues.py` is the one where I broke the rules on purpose. `fortune_teller_fixed.py` is the same program but written the way we were supposed to. They both run, they just ask the questions differently. The bad one makes you type your month and lucky number three times which is annoying, the good one only asks once.

To run them: `python fortune_teller_issues.py` or `python fortune_teller_fixed.py`

The 3 principles I picked

We had to pick 3 from the list. I went with DRY, Single Responsibility, and Document your Code because those were the easiest to show in a small program without making a bunch of classes.

- **DRY (Don't Repeat Yourself)** - basically don't copy paste the same code into a bunch of functions. If you have to fix it you don't want to fix it in 3 places.
- **Single responsibility** - each function should do one thing. Like getting input shouldn't also print the fortune.
- **Document your code** - comments should actually help. The lecture talked about bad comments too, like comments that just repeat the line, comments that are wrong, and leaving old code commented out.

DRY

Bad version

In the bad file I have `love_fortune`, `job_fortune`, and `luck_fortune`. I wrote `love_fortune` first and then I copied it twice and just changed the fortune strings. So the giant `if/elif` for the months is in there 3 times. `luck_fortune` even has a slightly different prompt ("what month were you born?") because I edited it after pasting and didn't make the other two match.

If I wanted to let people type "sept" for September I'd have to change all 3 functions. That's the whole point of DRY. Also I imported `random` at the top and never even used it because I was going to do `random.choice` and then used `% 4` instead and forgot to delete the import.

```
def love_fortune():
    month = input("Birth month (1-12 or name): ")
    month = month.strip().lower()
    if month == "1" or month == "january":
        m = 1
    elif month == "2" or month == "february":
        m = 2
    # ... this goes on for every month
    # and then job_fortune() has the exact same thing
    # and luck_fortune() has it again
```

The month if-else that I pasted 3 times.

Good version

In the good file the months are in a dictionary so I only have to write them once.

month_to_number() is the only place that converts the input. pick_fortune() is the only place that picks from a list. Then make_fortunes() just calls pick_fortune three times with the 3 different lists. If I add another fortune category I just make another list, I don't copy a whole function.

```
def month_to_number(text):
    text = text.strip().lower()
    if text in months:
        return months[text]
    if text.isdigit():
        n = int(text)
        if n >= 1 and n <= 12:
            return n
    return None

def pick_fortune(fortunes, seed):
    # % so if you put the same month + lucky number you get the same fortune again
    return fortunes[seed % len(fortunes)]

def make_fortunes(month, lucky):
    seed = month + lucky
    love = pick_fortune(love_fortunes, seed)
    career = pick_fortune(career_fortunes, seed)
    luck = pick_fortune(luck_fortunes, seed)
    return love, career, luck
```

Same logic, but it's only written once.

Single Responsibility

Bad version

get_name() is a mess. It prints the welcome banner AND asks for the name AND saves it to a global. love_fortune does like 4 jobs: it asks for the month, asks for the lucky number, picks a fortune, and prints it. I also used globals (n, m, num) so the functions are all tied together. You can't really tell what anything does without reading the whole file.

```
def get_name():
    global n
    print(" FORTUNE TELLER") # printing
    x = input("What is your name? ") # input
    n = x # global
    return x

def main():
    get_name()
    love_fortune() # asks stuff, picks, AND prints
    job_fortune()
    luck_fortune()
```

Good version

I split it up so each function has one job. `get_name` / `get_month` / `get_lucky_number` only get input. `month_to_number` only converts the month (it doesn't print anything). `pick_fortune` only picks. `get_info()` just calls the input functions. `make_fortunes()` only picks the 3 fortunes. `print_fortunes()` only prints. `main()` is just those 3 steps in order.

```
def main():
    name, month, lucky = get_info()
    love, career, luck = make_fortunes(month, lucky)
    print_fortunes(name, love, career, luck)
```

That's 9 functions which is more than the 3 we needed. I know some of the input ones are small but that's kind of the point, they're not doing extra stuff.

Document your Code

This one was about showing the bad commenting habits from class, not just "I added comments."

Bad comments in the issues file

- **Comments that just say what the next line does.** Like "# strip spaces" right above `x = x.strip()`. You can already read that.
- **Wrong comment.** `get_name` says "# returns the age" but it returns the name. I started from an old program I had that asked for a birthday and I forgot to change that comment. That's the stale comment thing from lecture.
- **Commented out code.** There's a `# print(x)` and `# i = i + 1` and a TODO save to a file that I never did. You're supposed to delete that, not leave it.
- **Missing a comment where it actually needed one.** `(m + num) % 4` is the part that's not obvious and I didn't explain it in the bad file.

```
def get_name():
    # this function gets the name
    x = input("What is your name? ")
    # strip spaces
    x = x.strip()
    # print(x)
    # i = i + 1
    n = x
    # returns the age
    return x
```

A bunch of comments that either don't help or are just wrong.

Good version

I put a short docstring on the functions that aren't totally obvious. I didn't comment `strip()` because that's self explanatory. The one comment I did add in `pick_fortune` is why I used `%` instead of `random`, so if you type the same month and number you get the same fortune. That's a why comment not a what comment.

```
def pick_fortune(fortunes, seed):
    # % so if you put the same month + lucky number you get the same fortune again
    return fortunes[seed % len(fortunes)]
```

Quick comparison

	issues.py (bad)	fixed.py (good)
DRY	Copied the month if-else into 3 functions	Dictionary + month_to_number + pick_fortune
SRP	love_fortune asks, picks, and prints. Globals everywhere.	get input, then make fortunes, then print. main just calls those.
Comments	"returns the age", # strip spaces, leftover print(x)	Docstrings on the functions + one comment explaining the %
When you run it	Asks month and lucky number 3 times	Asks each thing once

GitHub (bonus)

I put everything on github under my account. I made feature/violations for the bad file and feature/fixes for the good file and merged them with pull requests.

- Repo: <https://github.com/dsamanta12344/coding-best-practices-activity>
- PR 1 (bad version): <https://github.com/dsamanta12344/coding-best-practices-activity/pull/1>
- PR 2 (good version): <https://github.com/dsamanta12344/coding-best-practices-activity/pull/2>

Screenshots of the repo, the branches, the commits, and both PRs are below.

The screenshot shows the GitHub repository page for 'dsamanta12344/coding-best-practices-activity'. The repository is public and has 3 branches and 0 tags. The main branch is selected. The repository contains the following files:

- `.gitignore`: Add assignment README and gitignore (3 minutes ago)
- `README.md`: Add assignment README and gitignore (3 minutes ago)
- `fortune_teller_fixed.py`: Add fortune teller that follows DRY, SRP, and documentation (3 minutes ago)
- `fortune_teller_issues.py`: Add fortune teller that violates DRY, SRP, and documentation (3 minutes ago)

The README file is open, showing the title 'Coding Best Practices Activity' and the following text:

A pair of command-line fortune tellers used to demonstrate three programming principles:

1. **DRY** (Don't Repeat Yourself)
2. **Single Responsibility**
3. **Document your code** (including common commenting errors)

The 'Files' section shows a table with the following content:

File	Role
fortune_teller_issues.py	Same program, written to violate the three principles

The right sidebar shows the repository's statistics: 0 stars, 0 watching, 0 forks, and 0 releases. The contributors section lists 'dsamanta12344' (Deb Samanta) as the sole contributor. The languages section shows Python at 100%.

My github repo

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Branches

Overview Active Stale All

Search branches...

Default

Branch	Updated	Check status	Behind / Ahead	Pull request
main	now		Default	

Active branches

Branch	Updated	Check status	Behind / Ahead	Pull request
feature/fixes	1 minute ago		3 0	#2
feature/violations	1 minute ago		3 0	#1

The 3 branches (main, feature/fixes, feature/violations)

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Commits

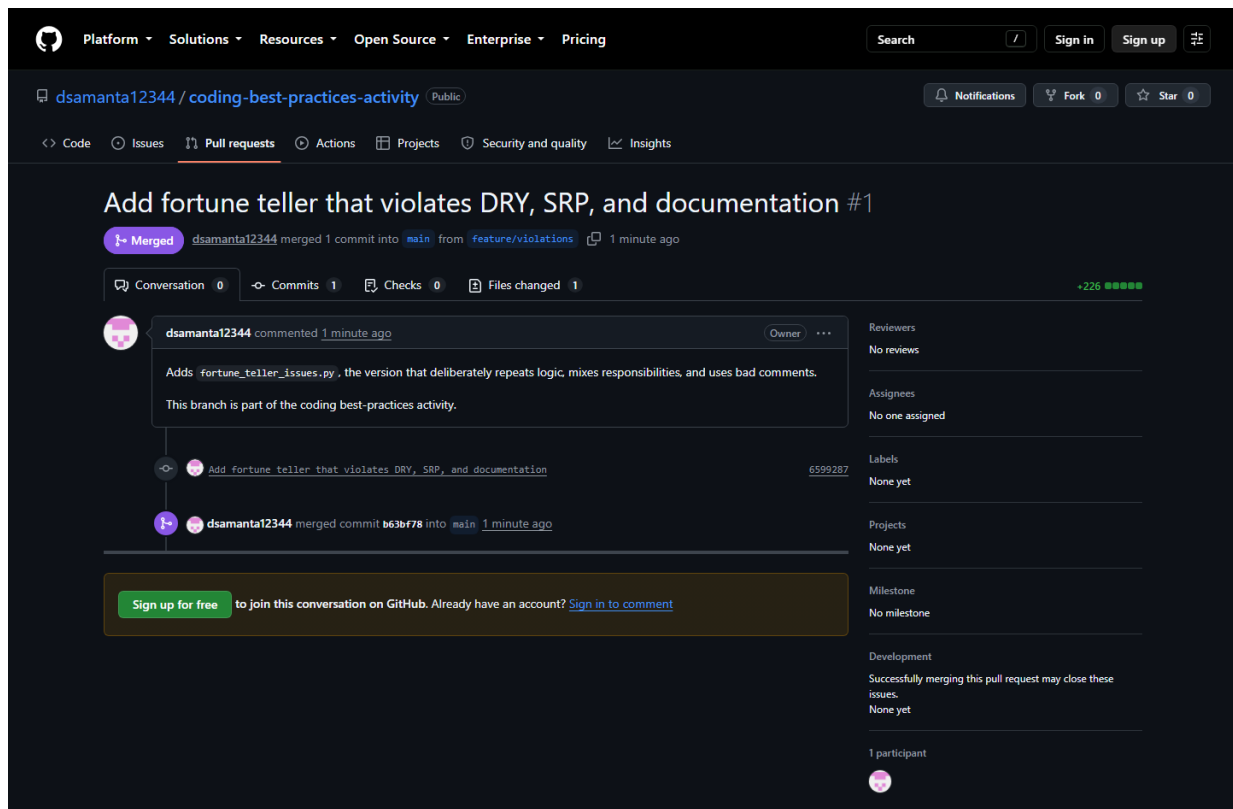
main

All users All time

Commits on Sep 18, 2026

- Merge pull request #2
dsamanta12344 authored 1 minute ago
Verified cefcaf
- Merge pull request #1
dsamanta12344 authored 1 minute ago
Verified b63bf78
- Add fortune teller that follows DRY, SRP, and documentation
dsamanta12344 committed 3 minutes ago
4c6cda7
- Add fortune teller that violates DRY, SRP, and documentation
dsamanta12344 committed 3 minutes ago
6599287
- Add assignment README and gitignore
dsamanta12344 committed 3 minutes ago
05fd678

Commit history



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Notifications Fork 0 Star 0

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Add fortune teller that violates DRY, SRP, and documentation #1

Merged dsamanta12344 merged 1 commit into main from feature/feature 1 minute ago

Conversation 0 Commits 1 Checks 0 Files changed 1 +226

dsamanta12344 commented 1 minute ago

Adds fortune_teller_issues.py, the version that deliberately repeats logic, mixes responsibilities, and uses bad comments.

This branch is part of the coding best-practices activity.

6599287

dsamanta12344 merged commit b63bf78 into main 1 minute ago

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Reviewers
No reviews

Assignees
No one assigned

Labels
None yet

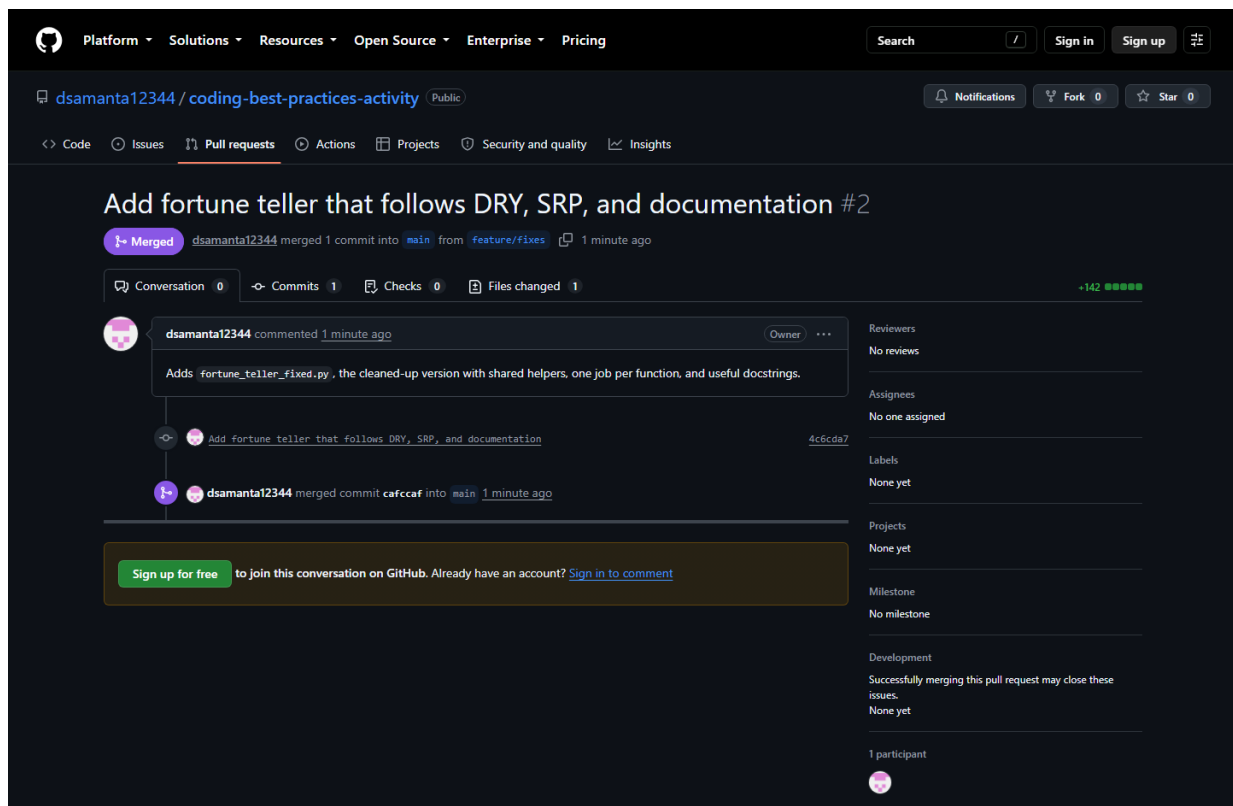
Projects
None yet

Milestone
No milestone

Development
Successfully merging this pull request may close these issues.
None yet

1 participant

Pull request for the bad version, already merged



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Add fortune teller that follows DRY, SRP, and documentation #2

Merged dsamanta12344 merged 1 commit into main from feature/feature 1 minute ago

Conversation 0 Commits 1 Checks 0 Files changed 1 +142

dsamanta12344 commented 1 minute ago

Adds fortune_teller_fixed.py, the cleaned-up version with shared helpers, one job per function, and useful docstrings.

4c6cda7

dsamanta12344 merged commit cafccaf into main 1 minute ago

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Reviewers
No reviews

Assignees
No one assigned

Labels
None yet

Projects
None yet

Milestone
No milestone

Development
Successfully merging this pull request may close these issues.
None yet

1 participant

Pull request for the good version, already merged